Sprint 1 Timeline Development

Deliverable 4: User Authentication | Source: Sprint 1 Work Breakdown Structure (WBS v1.1.0) | Version v1.0.0

PERT Estimation Technique

Each Sprint 1 task from the Sprint 1 Work Breakdown Structure is estimated using the Program Evaluation Review Technique (PERT). Three time estimates are gathered per task — Optimistic (O), Most Likely (M), and Pessimistic (P), each in working days and each set in increments of 0.25 — and combined into a single Expected Time (Te) using the weighted formula:

Te = (O + 4M + P) / 6

This weighs the most likely estimate four times more heavily than the best- and worst-case scenarios, producing a realistic single duration for planning purposes. Te is rounded to the nearest 0.25 day.

PERT Estimation Table (Sprint 1 Tasks)

Code	Work Package	O	M	P	Te = (O+4M+P)/6
1.0	**Sprint 1 Documents**				**Σ 6.00 days**
1.1	Sprint 1 WBS (PDF)	0.5	1	1.5	1.00 days
1.2	Sprint 1 PERT Timeline (PDF)	0.5	1	1.5	1.00 days
1.3	Backend Test Matrix (PDF)	1	2	3	2.00 days
1.4	Frontend Test Matrix (PDF)	1	2	3	2.00 days
2.0	**Registration**				**Σ 10.00 days**
2.1.1	Users Table Migration	0.5	1	1.5	1.00 days
2.1.2	Projects Table Migration (minimal, for Inbox)	0.25	0.5	1	0.50 days
2.2.1	Account Creation API (Password Hashing, Input Validation)	1	2	3	2.00 days
2.2.2	Username & Email Availability Check	0.5	1	1.5	1.00 days
2.2.3	Default Inbox Project Creation	0.5	1	1.5	1.00 days
2.3.1	Sign-Up UI	1.5	2.5	4	2.50 days
2.4.1	Registration Backend Integration Tests (API + Supabase)	0.5	1	2	1.00 days
2.4.2	Registration Frontend Integration Tests (Sign-Up UI + API)	0.5	1	2	1.00 days
3.0	**Login & Session**				**Σ 8.50 days**
3.2.1	Login API (Credential Verification, Input Validation)	1	1.5	2.5	1.50 days
3.2.2	Session Token Handling (Token Utilities)	1	1.5	2.5	1.50 days
3.3.1	Login UI	1	2	3	2.00 days
3.3.2	Session Restore	0.5	1	2	1.00 days

Code	Work Package	O	M	P	Te = (O+4M+P)/6
3.3.3	Logout	0.25	0.5	1	0.50 days
3.4.1	Login & Session Backend Integration Tests (API + Supabase)	0.5	1	1.5	1.00 days
3.4.2	Login & Session Frontend Integration Tests (Login, Session Restore, Logout + API)	0.5	1	2	1.00 days
4.0	**Password Recovery**				**Σ 11.00 days**
4.1.1	Password Reset Table Migration	0.5	0.75	1	0.75 days
4.2.1	Forgot Password API (Input Validation)	0.5	1	1.5	1.00 days
4.2.2	Reset Link Email	1	1.5	3	1.75 days
4.2.3	Reset Token Validation API	0.5	1	1.5	1.00 days
4.2.4	Reset Password API (Password Hashing, Input Validation)	1	1.5	2.5	1.50 days
4.3.1	Forgot Password UI	0.5	1	2	1.00 days
4.3.2	Reset Password UI	1	1.5	2.5	1.50 days
4.4.1	Password Recovery Backend Integration Tests (API + Supabase + Email)	1	1.5	2.5	1.50 days
4.4.2	Password Recovery Frontend Integration Tests (Forgot/Reset UI + API)	0.5	1	2	1.00 days
5.0	**Profile Management**				**Σ 4.50 days**
5.2.1	Update Profile API (Input Validation)	0.5	1	2	1.00 days
5.3.1	Profile Settings UI	1	1.5	2.5	1.50 days
5.4.1	Profile Management Backend Integration Tests (API + Supabase)	0.5	1	1.5	1.00 days
5.4.2	Profile Management Frontend Integration Tests (Profile Settings UI + API)	0.5	1	1.5	1.00 days
6.0	**Sprint 1 System Tests**				**Σ 2.50 days**
6.1	Sprint 1 End-to-End Tests (Registration, Login & Session, Password Recovery, Profile Management in one browser flow)	1.5	2.5	4	2.50 days

Grand Total (Sprint 1): Σ 42.50 person-days of estimated effort.

Notes

- Tasks 3.1.1 (Login & Session) and 5.1.1 (Profile Management) require no new Supabase migration and carry no estimate, since they reuse the Users Table built in 2.1.1.
- Task 2.1.2 (Projects Table Migration) is included so Sprint 1 can stand alone. If the projects table is built earlier, remove it and its estimate.
- Task 1.2 is this document. Tasks 1.3 and 1.4 are written first with every case set to Failing; the integration test tasks (2.4, 3.4, 4.4, 5.4) and the end-to-end task (6.1) include updating the matrices' Status column to Passing.
- Estimates are person-days of effort, summed across tasks. They are not calendar duration, since backend and frontend work can overlap.